Delivering oxygen with nanobubbles to retinal cells

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Background: Nanobubbles are small, < 1 μm, gaseous bubbles. Recent research has shown that they persist in solution for extended time periods, even for days, although theoretically, they should dissolve on the timescale of milliseconds. Here, we investigate whether nanobubbles participate in gas exchange in isolated tissue. We use the retina as a biosensor, since it is among the metabolically most active tissues. Without an additional oxygen supply, high metabolism results in hypoxia, which quickly reduces retinal signalling.

Methods: A custom-made microfluidics system was used to produce oxygen and carbogen nanobubbles. The nanobubble size distribution was determined with dynamic light scattering, and the oxygen concentration was measured with the Oxygraph+ system. The effect of nanobubbles on the retina was studied with transretinal ERG using WT C57BL/6J mice. The dark-adapted retina was stimulated with 1 ms 530 nm full-field flashes. Ames' medium was prepared from the water in which the nanobubbles had been produced the previous day. HEPES was used as a pH-buffer, as no additional bubbling was used in the solutions containing nanobubbles. 100 μM $BaCl_2$ was used to block the glial component, and in some experiments, 10 mM bicarbonate was added to the solution.

Results: The oxygen concentration in nanobubble dispersions was lower than in solutions constantly bubbled with traditional microbubblers but higher than in air-saturated solutions. The a-wave viability was sustained in nanobubble dispersions for several hours. The b-wave amplitude decreased in nanobubble dispersions but remained stable for several hours. The addition of bicarbonate increased both a-wave and b-wave viability.

Conclusions: The amount of oxygen achieved with this nanobubble production system maintains the a-wave, but the b-wave seems to require higher oxygen concentrations. The nanobubble production system should be developed further to achieve higher oxygen concentrations. Bicarbonate may have a role in cellular viability, not only by pH buffering but also with other mechanisms.

Keywords: ERG, transretinal ERG, nanobubble, retinal viability